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中药诱导毒性生化生物标志物：综述更新

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文章信息

关键词：传统中医 植物化学成分 副作用 毒理生物标志物

摘要

民族药理学相关性：传统中医（TCM）在中国被广泛用于疾病治疗，并因其高疗效和低副作用风险而成为药物开发的重要资源。然而，越来越多的毒性报告引起了全球的广泛关注。应对中医药引起的毒性的一大挑战是缺乏用于诊断和预测其毒性的特异性和敏感性生物标志物。鉴定反映中医药诱导毒性的毒理学生物标志物对于及时检测和干预至关重要，并为阐明潜在的毒性机制和关键靶点提供了重要线索。

研究目的：本文旨在总结和分类一些中药及其所含植物化学成分可能引起的副作用的毒理学生物标志物。

方法：使用关键词“生物标志物”、“中医药”、“中草药”、“植物化学成分”、“天然产物”、“毒性”、“肝毒性”、“肾毒性”、“心脏毒性”从文献数据库（包括PubMed, Web of Science）中收集截至2024年10月的相关信息。

结果：研究表明，需要更灵敏和更特异的生物标志物来反映中药的不良反应。PA-蛋白加合物和AA-DNA加合物可作为含PA和AA的中药引起的肝毒性和肾毒性的诊断性生物标志物。多种miRNA，如miRNA-122-3p、miRNA-5099和miRNA-21-3p，以及一些内源性代谢物，如次黄嘌呤、胆碱和L-缬氨酸，可能作为与中药引起的肝毒性、肾毒性和心脏毒性相关的潜在生物标志物。

结论：在本综述中，不同的研究表明，DNA/蛋白质加合物、非编码RNA、内源性代谢物等显示出作为中药诱导的毒性新早期预警生物标志物的潜力，且具有高度特异性和敏感性。

1. 介绍

中医（TCM）是一种常见的治疗方法，用于在中国的各种疾病中，它因其疗效显著且副作用小而受到重视（Tong 和 Deng，2020）。然而，由于使用不当，越来越多的报告显示中医药引发的不良反应。

缩略语：AA，马兜铃酸；ADR，不良药物反应；ALT，丙氨酸氨基转移酶；AST，天冬氨酸氨基转移酶；BUN，血尿素氮；BNP，脑钠肽；CFDA，中国食品药品监督管理局；CHE，胆碱酯酶；CYP，细胞色素P450蛋白；DHPA，脱氢吡咯里西啶生物碱；DIKI，药物性肾损伤；DILI，药物性肝损伤；DV，异叶金花茶；HSOS，肝窦阻塞综合征；KIM-1，肾损伤分子-1；lncRNA，长链非编码RNA；MCP-1，单核细胞趋化蛋白-1；miRNA，微小RNA；PA，吡咯里西啶生物碱；PMR，何首乌根；SQLE，角鲨烯合酶；TCM，中医药；TNFα，肿瘤坏死因子α；TSG，2,3,5,4′-四羟基二苯乙烯；TSN，土茛若素；VEGF，血管内皮生长因子。

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Biochemical biomarkers for the toxicity induced by Traditional Chinese Medicine: A review update

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ABSTRACT

Ethnopharmacological relevance: Traditional Chinese medicine (TCM) is widely used in China for disease treatment and has become a valuable resource for drug development due to its high efficacy and low risk of side-effects. However, growing toxicity reports have garnered significant global attention. A major challenge in addressing TCM-induced toxicity is lack of specific and sensitive biomarkers for diagnosing and predicting its toxicity. Identifying toxicological biomarkers reflecting TCM-induced toxicity is crucial for timely detection and intervention, and provides significant clues for elucidating the underlying toxic mechanism and key target.

Aim of the study: This article aims to summarize and classify some potential toxicological biomarkers for side-effects induced by TCM and its contained phytochemical ingredients.

Methods: The keywords “biomarkers”, “traditional Chinese medicine”, “Chinese herb”, “phytochemical ingredient”, “natural product”, “toxicity”, “hepatotoxicity”, “nephrotoxicity”, “cardiotoxicity” were used to collect relevant information from literature databases (including PubMed, Web of Science) up to October 2024.

Results: Research has indicated that more sensitive and specific biomarkers are needed for reflecting TCM’s side-effects. PA-protein adducts and AA-DNA adducts could be served as diagnostic biomarkers for hepatotoxicity and nephrotoxicity induced by TCM containing PA and AA, respectively. Multiple miRNAs like miRNA-122-3p, miRNA-5099, and miRNA-21-3p, as well as some endogenous metabolites such as hypoxanthine, choline, and L-valine could be potential biomarkers associated with TCM-induced hepatotoxicity, nephrotoxicity, and cardiotoxicity.

Conclusion: In this review, different research demonstrates that DNA/protein-adducts, noncoding RNAs, endogenous metabolites and so on show the potential to be new early-warning biomarkers for TCM-induced toxicity with high specificity and sensitivity.

1. Introduction

Traditional Chinese medicine (TCM) is a prevalent treatment for various diseases in China, valued for its therapeutic efficacy and minimal side-effects (Tong and Deng, 2020). However, increasing reports of adverse reactions induced by TCM stemming from improper use,

Abbreviations: AA, aristolochic acid; ADR, adverse drug reaction; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BUN, blood urea nitrogen; BNP, brain natriuretic peptide; CFDA, China Food and Drug Administration; CHE, Cholinesterase; CYP, Cytochrome P450 proteins; DHPA, dehydro-pyrrolizidine alkaloid; DIKI, drug-induced kidney injury; DILI, drug-induced liver injury; DV, *Dysosma versipellis*; HSOS, hepatic sinusoidal obstruction syndrome; KIM-1, kidney injury molecule-1; lncRNA, long non-coding RNA; MCP-1, monocyte chemotactic protein-1; miRNA, MicroRNA; PA, pyrrolizidine alkaloid; PMR, Polygonum multiflorum Radix; SQLE, Squalene synthase; TCM, Traditional Chinese Medicine; TNFα, tumor necrosis factor alpha; TSG, 2,3,5,4′-tetrahydroxyl-diphenylethylene; TSN, toosendanin; VEGF, vascular endothelial growth factor.

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污染或固有毒性已经引起了重大关注。这些问题不仅影响中医药在临床中的安全和有效使用，也阻碍了进一步的药物开发和国际认可（Chan 等, 2015；Claure-Del 和 Espinosa-Cuevas, 2021；Teschke 等, 2021）。根据中国国家药品不良反应（ADR）监测年度报告，中国与ADR相关的病例总数为202万例，其中中医药占12.8%。据报道，中医药的摄入通常与心脏毒性、肾脏毒性、神经毒性，尤其是肝毒性有关（Chen 等, 2011）。以中药的肝毒性为例，2019年的一项全国性回顾性研究表明，在25,927例被诊断为药物性肝损伤（DILI）的患者中，26.81%的病例与中医药相关（Shen 等, 2019；Li 等, 2022）。中医药的复杂性，包括多组分天然物

生化标志物反映了细胞和器官生物功能的变化，可以指示疾病的存在、进展或改善（Schomaker 等，2019）。自20世纪90年代以来，越来越多的研究人员致力于探索生物标志物，因为它们在促进和加快获取药物有效性和安全性信息方面起着重要作用（Cohen 等，2015）。为了尽可能避免不良反应的发生，并确保中药在临床中的安全使用，尽早识别潜在损伤在中药引起的病理损害发生之前具有重要意义。因此，发现中药毒性相关的生物标志物有助于减少其潜在的不良反应，指导其安全有效的临床用药，并为中药解毒提供关键线索。然而，目前中药毒性研究的关键问题在于缺乏特异性和敏感性强的标志物

近年来，许多毒理学研究人员集中于发现具有高特异性、高敏感性和良好稳定性的生化标志物，以用于中药引起的毒性。在这些研究中，由有毒物质形成的蛋白质或DNA加合物已成为中药诱导毒性的有前景的生物标志物。例如，脱氢吡咯里西啶生物碱（DHPA）-蛋白加合物已成功被应用于吡咯里西啶生物碱（PA）引起的肝窦阻塞综合征（HSOS）的临床诊断，成为南京标准的一部分（Zhuge等，2019）。此外，非编码RNAs以其高敏感性而被报道为中药诱导毒性的潜在生物标志物。此外，包括细胞因子和酶在内的内源性代谢物和蛋白质，不仅能反映疾病的发生，还直接参与其进程，在监测中药诱导的毒性方面具有巨大潜力。

2. 中医引起毒性的经典生物标志物

经典的毒性生物标志物通常是指那些在临床实践中已被广泛使用数十年的指标，用于评估药物引起的毒性或监测疾病状态，它们也常被用来反映中药引起的毒性。如表1所总结的，经典生物标志物包括但不限于肝脏

表1 中医药诱导毒性经典生物标志物。

类型毒性	经典生物标志物	物体	参考
肝毒性	丙氨酸氨基转移酶	血清	高等人（2023）
	天冬氨酸氨基转移酶	血清	
	碱性磷酸酶	血清	
	总胆红素	血清	
心脏毒性	心脏肌钙蛋白T	血清	王等人（2019）
	心肌肌钙蛋白I	血清	
	B型利钠肽	血清	
	N端前b型利钠钛	血清	
神经毒性	神经特异性烯醇化酶	血清	Korfias 等人（2006）；
	S100 蛋白	血清	连等人（2023）
肾毒性	血尿素氮	血清	Gao 等人（2023）；Sun
	肌酐	血清	等人（2011）

用于评估肝毒性的酶如ALT和AST（Gao等，2023），以及用于检测中药引起的心脏毒性的心肌肌钙蛋白和脑钠肽（BNP）（Wang等，2019）。此外，BUN和血清肌酐被用于评估中药引起的肾毒性（Sun等，2011）。这些生物标志物主要从血清或尿液中采集，具有易于检测的特点。然而，由于中药成分的复杂性，这些传统生物标志物往往难以准确评估中药的特定毒性特征。此外，它们在特异性和敏感性方面常存在局限性，从而降低了区分中药引起的毒性与其他类型毒性或潜在疾病的有效性。因此，尽管这些经典生物标志物仍是有价值的工具，但它们的局限性凸显了需要更先进的生物标志物来反映中药独特的毒理学效应。

随着医学研究和实验技术的进步，生物标志物的探索达到了新的深度，导致发现了与中药诱导的毒性相关的新型生物标志物。这些新兴的生物标志物在准确反映中药毒性特征方面显示出更大的潜力。

3. 蛋白质或DNA加合物作为中药诱导毒性的生物标志物

先前的研究报道称，药物-蛋白/DNA 加合物被认为是药理学或毒理学指标的重要特征，并且可以被轻松检测到（Gan 等，2016）。研究已经表明，对乙酰氨基酚-蛋白加合物可作为临床中监测对乙酰氨基酚引起的急性肝损伤的早期生物标志物（Chiew 等，2020；James 等, 2009）。同时，DNA 加合物也被认为是疾病的生物标志物（Hernandez-Castillo 等，2020）。因此，蛋白或 DNA 加合物在作为中药引起的毒性生物标志物方面具有巨大潜力。

3.1. 蛋白质加合物作为生物标志物

吡咯里西啶生物碱（PA）是一种天然的肝毒性生物碱，广泛存在于菊科、紫草科、豆科和兰科等植物中，其中许多植物被列入中国药典或常用作民间药物，如日本菊苣、爬山虎属植物、红根草等（Chojkier, 2003；Zhang 等, 2008；Luo 等, 2019）。临床报告发现，含有PA的草药可引起急性肝毒性，其典型病理症状包括肝肿大、黄疸和腹水，被称为肝静脉闭塞性疾病（HSOS）（DeLeve 等, 1999；Chojkier, 2003；Zhu 等, 2021a）。研究发现，肝脏细胞色素P450可催化PA生成DHPA，后者可进一步共价结合于RNA、DNA或蛋白等大分子，从而导致毒性（McLean, 1970；Fu 等, 2004；Li 等, 2011；Fu, 2017）。

contamination, or inherent toxicity has raised significant concerns. These issues not only affect the safe and effective usage of TCM in clinic but also impede further drug development and international acceptance (Chan et al., 2015; Claure-Del and Espinosa-Cuevas, 2021; Teschke et al., 2021). As described in the annual report of national adverse drug reaction (ADR) monitoring by China Food and Drug Administration (CFDA), there were a total of 2.02 million cases related to ADR in China, of which TCM contributed to 12.8%. It was reported that the intake of TCM is usually in connection with cardiotoxicity, nephrotoxicity, neurotoxicity, and especially hepatotoxicity (Chen et al., 2011). Taking the hepatotoxicity of TCM as an example, a nationwide retrospective study in 2019 suggested that 26.81% of the cases were related to TCM in 25,927 patients diagnosed with drug-induced liver injury (DILI) (Shen et al., 2019; Li et al., 2022). The complexity of TCM, including the multi-component nature and diverse pharmacological interactions, contributes to the difficulty in diagnosing and managing its toxicity. Furthermore, the multifaceted and often delayed manifestations of TCM-induced toxicity, which may affect multiple organs, pose challenges for timely and accurate diagnosis, thereby limiting TCM's clinical application and global advancement. Hence, it is crucial to identify the key molecules in TCM-induced toxicity and develop proper diagnostic approaches.

Biochemical markers, which reflect changes in the biological functions of cells and organs, can indicate the presence, progression, or improvement of diseases (Schomaker et al., 2019). Since the 1990s, more and more researchers have devoted to exploring biomarkers due to their role in facilitating and accelerating the access to obtain the information about drug efficacy and safety (Cohen et al., 2015). In order to avoid the occurrence of side-effects as much as possible and ensure the safe use of TCM in clinic, it is of great significance to identify the potential injury as soon as possible before the occurrence of pathological damages induced by TCM. Therefore, the discovery of biomarkers for toxicity induced by TCM will be helpful to reduce their potential ADR and guide their clinical medication safely and effectively, and it will also provide key clues for its detoxification. However, the current key problem for toxicity induced by TCM is the shortage of specific, sensitive and accessible biomarkers for diagnosing and predicting its toxicity, and many classic biomarkers, such as alanine aminotransferase (ALT), aspartate amino transferase (AST), blood urea nitrogen (BUN) and so on, are all insensitive and nonspecific (Griffin et al., 2019).

In recent decades, many toxicological researchers focused on discovering biochemical markers with the characteristics of high specificity, high sensitivity, and good stability for toxicity induced by TCM. Among these studies, protein or DNA adducts formed by toxic substances have emerged as promising biomarkers for TCM-induced toxicity. For instance, dehydro-pyrrolizidine alkaloid (DHPA)-protein adducts have been successfully incorporated into clinical diagnostics for pyrrolizidine alkaloid (PA)-induced hepatic sinusoidal obstruction syndrome (HSOS), forming part of the Nanjing criteria (Zhuge et al., 2019). Additionally, non-coding RNAs, known for their high sensitivity, have been reported as potential biomarkers for TCM-induced toxicity. Furthermore, endogenous metabolites and proteins, including cytokines and enzymes, which not only reflect the occurrence of disease but also directly participate in its progression, hold great potential as biomarkers for monitoring TCM-induced toxicity. This review aims to comprehensively summarize the biomarkers identified to date for TCM-induced toxicity, providing valuable insights to ensure the safe and effective clinical application of TCM.

2. Classic biomarkers for toxicity induced by TCM

Classic biomarkers of toxicity typically refer to those indicators that have been widely used in clinical practice for decades to evaluate drug-induced toxicity or monitor disease status, and they have also been commonly utilized to reflect TCM-induced toxicity. As summarized in Table 1, classical biomarkers include but are not limited to, liver

Table 1
Classic biomarkers for toxicity induced by TCM.

Type of toxicity	Classic biomarker	Object	Reference
Hepatoxicity	Alanine aminotransferase	Serum	Gao et al. (2023)
	Aspartate aminotransferase	Serum	
	Alkaline phosphatase	Serum	
	Total bilirubin	Serum	
Cardiotoxicity	Cardiac troponin T	Serum	Wang et al. (2019)
	Cardiac troponin I	Serum	
	B-type natriuretic peptide	Serum	
	N-terminal pre-b-type natriuretic Titanium	Serum	
Neurotoxicity	Neuro-specific enolase	Serum	Korfias et al. (2006);
	S100 protein	Serum	Lian et al. (2023)
Nephrotoxicity	Blood urea nitrogen	Serum	Gao et al. (2023); Sun
	Creatinine	Serum	et al. (2011)

enzymes like ALT and AST for assessing hepatotoxicity (Gao et al., 2023), and cardiac troponin and brain natriuretic peptide (BNP) for detecting cardiotoxicity caused by TCM (Wang et al., 2019). In addition, BUN and serum creatinine were utilized to assess nephrotoxicity induced by TCM (Sun et al., 2011). These biomarkers are mainly collected from serum or urine, and they have the characteristics of easy detection. However, due to the complexity of TCM components, these conventional biomarkers often struggle to accurately assess the specific toxicity characteristics of TCM. Moreover, they often exhibit limitations in terms of specificity and sensitivity, reducing their effectiveness in distinguishing TCM-induced toxicity from other forms of toxicity or underlying conditions. Thus, while these classical biomarkers remain valuable tools, their limitations highlight the need for more advanced biomarkers to reflect the unique toxicological effects of TCM.

With the advancement of medical research and experimental technologies, the exploration of biomarkers has reached new depths, leading to the discovery of novel biomarkers associated with toxicity induced by TCM. These emerging biomarkers show greater potential in accurately reflecting the toxicity characteristics of TCM.

3. Protein or DNA adducts as biomarkers for toxicity induced by TCM

Previous study has reported that drug-protein/DNA adducts were considered as an important feature of pharmacological or toxicological indicators, and they can be easily detected (Gan et al., 2016). Studies have already demonstrated that acetaminophen-protein adducts could be served as early biomarkers for monitoring acetaminophen-induced acute liver injury in clinic (Chiew et al., 2020; James et al., 2009). Meanwhile, DNA adducts have also been recognized as biomarkers for diseases (Hernandez-Castillo et al., 2020). Hence, protein or DNA adducts exhibit great potential to serve as biomarkers for toxicity induced by TCM.

3.1. Protein adducts as the biomarker

PA, a type of natural hepatotoxic alkaloid, is globally distributed in plants belong to Compositae, Boraginaceae, Leguminosae and Orchidaceae families, among which many plants are listed in China Pharmacopeia or commonly used as folk medicines, such as *Gynura japonica*, *Senecio scandens* Buch.-Ham., *Arnebia euchroma* (Royle) Johnst. et al. (Chojkier, 2003; Zhang et al., 2008; Luo et al., 2019). Clinical reports found that herbal medicines containing PA could induce acute liver toxicity with representative pathological symptoms including hepatomegaly, jaundice, and ascites, which is known as HSOS (DeLeve et al., 1999; Chojkier, 2003; Zhu et al., 2021a). Hepatic cytochrome P450 was discovered to catalyze PA into DHPA, which can further covalently bind to macromolecules like RNA, DNA or protein, causing toxicity (McLean, 1970; Fu et al., 2004; Li et al., 2011; Fu, 2017).

越来越多的报道表明，DHPA-蛋白加合物作为由PA诱导的肝毒性生物标志物具有重要潜力（Lu 等, 2018）。在PA诱导的肝损伤中，血清ALT酶活性与DHPA-蛋白加合物的形成之间存在高度显著的正相关，表明其作为PA诱导肝毒性生物标志物的潜在应用价值（Ma 等, 2018）。随后，在一位因误食含有肝毒性PA的民间草药小叶锦灯藤（Gynura japonica，土三七）而发生HSOS的患者血清中，鉴定出了DHPA-蛋白加合物（Lin 等, 2011）。此外，在一项包括40名患者的前瞻性研究中，其中23名患者为PA相关HSOS，17名患者为其他药物引起的肝损伤但未含PA，研究显示PA相关HSOS患者血液中DHPA-蛋白加合物的浓度范围为0.05至74.4 nM（Gao 等, 2015）。该研究表明，测量

在实验研究中，还在因含有吡咯里西啶生物碱（PA）的草药中毒的小鼠和大鼠体内发现了DHPA-蛋白加合物。例如，在给药关叶前旋花（Gynura japonica）后，发现小鼠和大鼠血液及肝脏中DHPA-蛋白加合物水平较高（Yang 等，2017a）。尽管目前尚无关于临床中其他含PA草药引起的肝毒性的报道，但发现三叶蛇根草（Crotalaria sessiliflora）提取物可在大鼠中诱导严重肝损伤，并且在肝组织中也检测到了DHPA-蛋白加合物（Ma 等，2021a）。这些研究结果表明，DHPA-蛋白加合物可作为监测含PA草药引起的肝损伤的特异性生物标志物。

最近，在43例PA中毒患者的血液中检测到一种DHPA-血红蛋白加合物，该加合物显示出明显更高的水平和更长的存在时间，提示其作为诊断PA含量草药引起的肝毒性的更可靠生物标志物的潜力（Ma等, 2021b）。同时，另一项研究显示，在PA中毒大鼠和摄入凤尾蕨患者的尿液样本中存在四种DHPA-氨基酸加合物，分别命名为吡咯-7-半胱氨酸、吡咯-9-半胱氨酸、吡咯-9-组氨酸和吡咯-7-乙酰半胱氨酸（Zhu等, 2021b）。这四种DHPA-氨基酸加合物表现出强大的潜力，可作为帮助诊断PA含草药引起的肝毒性的无创生物标志物。

除了PA，其他有毒中药中的有毒物质也可以形成蛋白加合物。从栀子果中分离出的主要活性天然化合物连翘苷可以被肠道微生物的β-葡萄糖苷酶水解为连翘素，研究发现连翘素可以与蛋白质中的L-赖氨酸残基相互作用，形成连翘素-蛋白加合物，从而在大鼠中引起肝损伤（Li等, 2019）。苦楝苷（TSN）是苦楝子中发现的主要肝毒性化合物，在《中国药典》中标注其毒性很低。先前的研究显示，TSN反应性代谢物的蛋白加合物可以在大鼠的肝脏和血液中检测到，并且这些蛋白加合物与TSN诱导的肝毒性严重程度呈良好相关（Zhuo等，2021）。传统草药土薯蓣（Dioscorea bulbifera L.）在临床实践中已被发现会引起肝损伤，实验结果也证实了其肝毒性（Ma

et al., 2014）。虎杖苷-B（Diosbulbin-B）是中华山药（Dioscorea bulbifera L）中的主要肝毒性化合物，可被肝脏细胞色素P450代谢生成其代谢物，该代谢物可进一步与蛋白质的赖氨酸残基结合，最终形成蛋白质加合物，这些加合物可以在血浆中检测到，并且与虎杖苷-B的剂量呈正相关（Hu et al., 2022）。然而，胡椒素-蛋白加合物（genipin-protein adducts）、TSN-蛋白加合物或虎杖苷-B-蛋白加合物是否可以作为其毒性的可靠生物标志物，仍需更多临床研究的证据。

3.2. DNA加合物作为生物标志物

研究还发现，在给予一系列肝毒性吡咯里西啶生物碱（PA）或膳食补充剂（如紫草或马蹄草根提取物、紫草复合油）或远志（Tussilago farfara L.，一种药用草本）的大鼠肝脏中存在DHPA-DNA加合物（Chou and Fu, 2006; He et al., 2017）。此外，研究表明，在给予PA的小鼠中，DHPA-DNA加合物表现出长期持久性，表明其作为反映PA引起的肝毒性的生物标志物具有巨大潜力（Zhu et al., 2017）。最近的一项研究进一步确认了肝脏DHPA-DNA加合物与血清DHPA-蛋白加合物之间存在良好的正相关（Zhu et al., 2022）。由于DHPA-蛋白加合物在血清中的丰度更高且更易检测，作者提出其比肝脏DHPA-DNA加合物更适合作为指示PA中毒的生物标志物。

马兜铃酸（AA）在各种中药中含量丰富，例如关木通（Aristolochia manshuriensis Kom.）、防己马兜铃（Aristolochia fangchi Y. C. Wu ex L. D. Chow et S. M. Hwang）、关木通（Aristolochia debilis Sieb. et Zucc）等。已有报道显示，AA 可引起严重的急性肾损伤和肾小管间质性疾病（Yang 等，2018；Jelakovic 等，2019；Baudoux 等，2022）。AA 可以转化为其代谢活性产物马兜铃内酯（aristololactam），该产物可共价结合 DNA 中的嘌呤形成马兜铃内酯-DNA 加合物，从而导致肾损伤，同时具有较强的致癌活性（Stiborova 等，2017；Anger 等，2020）。一项临床研究（Jelakovic 等，2012）显示，在 47 例 AA 诱发的肾病患者的肾皮质中检测到了马兜铃内酯-DNA 加合物。此外，马兜铃内酯-DNA 加合物显示出良好的敏感性和稳定性，并且对全基因组核苷酸切除修复具有抗性，因此具有成为可靠生物标志物的巨大潜力。

据报道，源自罗勒（Ocimum basilicum L.）和八角（Illicium verum Hook. f.）的茨烯可以被肝脏细胞色素P450代谢，其代谢产物可形成DNA加合物，这些加合物被认为是反映其肝毒性的潜在生物标志物（Rietjens 等，2005；Chen 等，2011）。此外，当茨烯与肝微粒体共孵育时，会形成DNA加合物和血红蛋白加合物（Bergau 等，2021）。然而，这些DNA或血红蛋白加合物在临床应用中仍需通过更可靠的实验和临床证据进一步验证。

以上所有研究都表明，这些蛋白质或DNA加合物（表2）（图1，以PA为例）可以作为中药及其活性成分所致毒性的高敏感性和高特异性诊断生物标志物，并且它们在临床上具有进一步应用的巨大潜力。

4. 非 编码RNA作为毒性诱导生物标志物 中医

非编码RNA，主要包括微小RNA（miRNA）和长非编码RNA（lncRNA），由于其高通量、高灵敏度且易于定量的特点，已越来越多地应用于毒理学生物标志物的研究（Joseph, 2017）。已有报道显示，一些miRNA已被用于癌症和心血管疾病的诊断与预后（Lujambio和Lowe, 2012；Wieczorek和Reszka, 2018；Zhou等, 2018）。因此，非编码RNA在中药诱导的毒性作为生物标志物方面具有重要潜力。

More and more reports have exhibited that DHPA-protein adducts hold significant promise as biomarkers for hepatotoxicity induced by PA (Lu et al., 2018). A highly significant positive correlation was found between serum ALT enzymatic activity and the formation of DHPA-protein adducts in liver injury induced by PA, indicating its potential application as a biomarker for hepatotoxicity induced by PA (Ma et al., 2018). Later, a DHPA-protein adduct was identified in the serum from a HSOS patient caused by the accidental ingestion of *Gynura japonica* (Tu-San-Qi) that is a folk herbal medicine containing hepatotoxic PA (Lin et al., 2011). Furthermore, a prospective study of forty patients, among whom 23 patients had PA-associated HSOS and 17 patients had liver injury caused by other drugs without PA, demonstrated that blood concentrations of DHPA-protein adducts ranged from 0.05 to 74.4 nM in PA-associated HSOS patients (Gao et al., 2015). This study demonstrated that the measurement of blood DHPA-protein adducts exhibited 100% sensitivity and 94.1% specificity in the diagnosis of PA-associated HSOS (Gao et al., 2015). Moreover, the levels of blood DHPA-protein adducts were found to be significantly correlated with the severity and clinical outcome of PA-related HSOS, because those patients who died or required liver transplantation within 6 months (categorized as the severe group) had significantly higher blood DHPA-protein adducts levels compared to those who fully recovered or developed chronic disease (categorized as the recovery/chronic disease group) (Gao et al., 2015). These clinical findings highlight the huge potential of blood DHPA-protein adducts as specific and sensitive biomarkers for assessing PA-induced hepatotoxicity. Furthermore, the detection of DHPA-protein adducts has already been integrated into clinical diagnostics for PA-induced HSOS, as outlined in the Nanjing criteria (Zhuge et al., 2019).

DHPA-protein adducts have also been found in experimental studies involving mice and rats intoxicated by herbal medicines containing PA. For example, a high level of DHPA-protein adducts was found in the blood and liver both in mice and rats after the administration of *Gynura japonica* (Yang et al., 2017a). Although there is still no report about the hepatotoxicity induced by other herbal medicines containing PA in clinic, the extract of *Crotalaria sessiliflora* was found to induce severe hepatic injury in rats and DHPA-protein adducts were also detected in liver tissues (Ma et al., 2021a). These findings suggest that DHPA-protein adducts could be utilized as specific biomarkers for monitoring liver injury induced by PA-containing herbal medicines.

Recently, a DHPA-hemoglobin adduct was identified in the blood from 43 patients with PA intoxication, and this adduct displayed obviously higher levels and longer persistence, suggesting its potential as a more reliable biomarker for diagnosing hepatotoxicity caused by PA-containing herbal medicines (Ma et al., 2021b). Meanwhile, another study manifested that there were four DHPA-amino acid adducts in urine samples from rats intoxicated with PA and patients who had ingested *Gynura japonica*, named pyrrole-7-cysteine, pyrrole-9-cysteine, pyrrole-9-histidine, and pyrrole-7-acetylcysteine (Zhu et al., 2021b). These four DHPA-amino acid adducts exhibited strong potential as non-invasive biomarkers for assisting in diagnosing hepatotoxicity induced by PA-containing herbal medicines.

Except PA, other toxic substances from toxic TCMs can also form protein adducts. Geniposide, a main active natural compound isolated from Gardeniae Fructus, can be hydrolyzed into genipin by intestinal microbiota β-glucosidase, and genipin was found to interact with L-lysine residue in the protein to form genipin-protein adducts, thereby inducing hepatic injury in rats (Li et al., 2019). Toosendanin (TSN) is the main hepatotoxic compound found in Toosendan Fructus, which is marked to have little toxicity in Chinese Pharmacopeia. A previous study manifested that protein adducts of the reactive metabolites of TSN could be found in the liver and blood from rats, and those protein adducts displayed a good correlation with the severity of TSN-induced hepatotoxicity (Zhuo et al., 2021). Traditional herbal medicine *Dioscorea bulbifera* L. has been found to cause hepatic damage during clinical practice, and experimental results also confirmed its hepatotoxicity (Ma et al., 2014). Diosbulbin-B, the primary hepatotoxic compound from *Dioscorea bulbifera* L, could be metabolized by hepatic cytochrome P450 to produce its metabolite that can further bind to lysine residues of protein and finally form protein adducts, which could be detected in the plasma and showed a positive relationship with doses of diosbulbin-B (Hu et al., 2022). However, whether genipin-protein adducts, TSN-protein adducts, or diosbulbin-B-protein adducts can be used as reliable biomarkers for their toxicity still requires more evidences from clinical investigation.

3.2. DNA adducts as the biomarker

DHPA-DNA adducts have also been discovered in the livers from rats given with a series of hepatotoxic PAs or dietary supplements (comfrey or coltsfoot root extract, comfrey compound oil) or *Tussilago farfara* L. (a medicinal herb) (Chou and Fu, 2006; He et al., 2017). Additionally, studies have shown that DHPA-DNA adducts exhibit long-term persistence in mice given with PA, indicating their huge potential as biomarkers for reflecting hepatotoxicity induced by PA (Zhu et al., 2017). A recent study further identified a good positive correlation between liver DHPA-DNA adducts and serum DHPA-protein adducts (Zhu et al., 2022). Since DHPA-protein adducts exhibit higher abundance in serum and are easier to be detected, the authors propose that it is more suitable than liver DHPA-DNA adducts as biomarkers for indicating PA intoxication.

Aristolochic acid (AA) is abundant in various TCMs, such as *Aristolochia manshuriensis* Kom., *Aristolochia fangchi* Y. C. Wu ex L. D. Chow et S. M. Hwang., *Aristolochia debilis* Sieb. et Zucc, and so on. AA was reported to induce severe acute renal injury and tubulointerstitial kidney disease (Yang et al., 2018; Jelakovic et al., 2019; Baudoux et al., 2022). AA could be converted into its metabolic active product aristololactam that covalently binds to purines in DNA to form aristolactam-DNA adducts, which can cause kidney injury and also exhibit strong carcinogenic activity (Stiborova et al., 2017; Anger et al., 2020). A clinical investigation (Jelakovic et al., 2012) manifested that aristolactam-DNA adducts were detected in renal cortex from 47 patients with nephrosis induced by AA. Moreover, aristolactam-DNA adducts displayed good sensitivity and stability, and they were resistant to global genomic nucleotide excision repair, so they have the huge potential to be reliable biomarkers for diagnosing nephropathy induced by AA-containing TCMs (Moriya et al., 2011; Jelakovic et al., 2012).

Estragole from *Ocimum basilicum* L. and *Illicium verum* Hook. f. have been reported to be metabolized by hepatic cytochrome P450, and its metabolites can form DNA-adducts that are considered potential biomarkers for reflecting their hepatotoxicity (Rietjens et al., 2005; Chen et al., 2011). Furthermore, DNA-adducts and hemoglobin-adduct were formed when estragole was incubated with hepatic microsomes (Bergau et al., 2021). However, the clinical application of those DNA- or hemoglobin-adducts requires further validation through more robust experimental and clinical evidences.

All those above studies imply that those protein- or DNA-adducts (Table 2) (Fig. 1, PA has been taken as an example) could be used as diagnostic biomarkers with high sensitivity and specificity for toxicity induced by TCM and its active ingredients, and they have the huge potential to be further applied in clinic.

4. Non-coding RNAs as biomarkers for toxicity induced by TCM

Non-coding RNAs, mainly including microRNA (miRNA) and long non-coding RNA (lncRNA), have increasingly been applied in the study of toxicological biomarkers due to their high throughput, high sensitivity, and easy to quantify (Joseph, 2017). It has been reported that some miRNAs have already been used for the diagnosis and prognosis of cancer and cardiovascular diseases (Lujambio and Lowe, 2012; Wieczorek and Reszka, 2018; Zhou et al., 2018). Hence, non-coding RNAs hold significant potential as biomarkers for toxicity induced by TCM.

表2 蛋白质/DNA加合物作为中药诱导毒性的生物标志物。

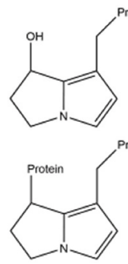
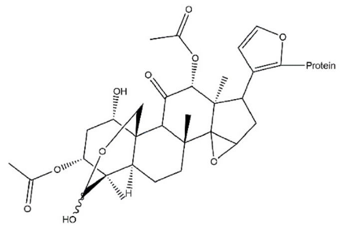
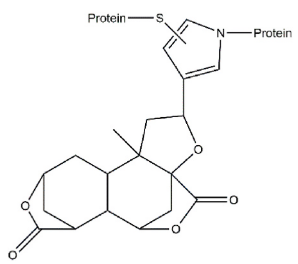
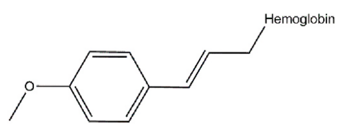
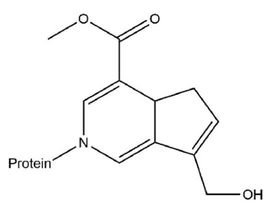
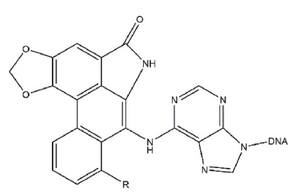
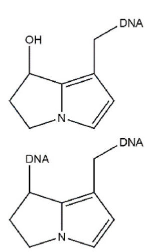
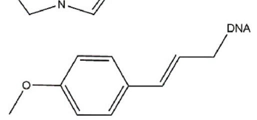
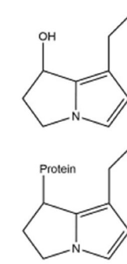
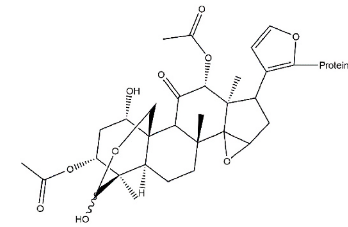
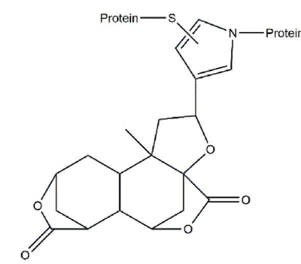
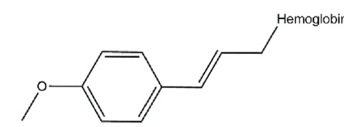
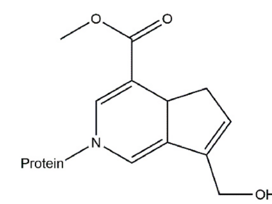
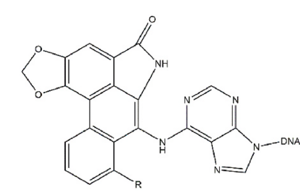
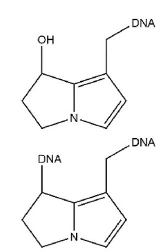
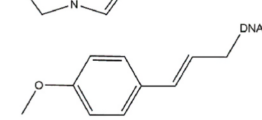
Type	天然产品	加合物结构	参考
蛋白质内收	吡咯里西啶生物碱		马等 (2018); 林等 (2011); 高等 (2015); 诸葛等 (2019); 马等 (2021a); 马 等 (2021b); 杨 等 (2017a)
	图森丹宁		卓等（2021）
	迪奥斯布尔宾-B		胡等人（2022年）
	豆蔻醛		Bergau 等（2021）
	连翘苷		李等人（2019年）
DNA加合物	马兜铃酸		Jelakovic 等（2012）；Stiborova 等（2017）；Anger 等（2020）
	吡咯里西啶生物碱		Zhu 等人（2017）；Zhu 等人（2022）
	豆蔻醛		Bergau 等（2021）

Table 2
Protein/DNA-Adducts as the biomarker for toxicity induced by TCM.

Type	Natural product	Adduct structure	Reference
Protein-adduct	Pyrrolizidine alkaloid		Ma et al. (2018); Lin et al. (2011); Gao et al. (2015); Zhuge et al. (2019); Ma et al. (2021a); Ma et al. (2021b); Yang et al. (2017a)
	Toosendanin		Zhuo et al. (2021)
	Diosbulbin-B		Hu et al. (2022)
	Estragole		Bergau et al. (2021)
	Geniposide		Li et al. (2019)
DNA-adduct	Aristolochic acid		Jelakovic et al. (2012); Stiborova et al. (2017); Anger et al. (2020)
	Pyrrolizidine alkaloid		Zhu et al. (2017); Zhu et al. (2022)
	Estragole		Bergau et al. (2021)

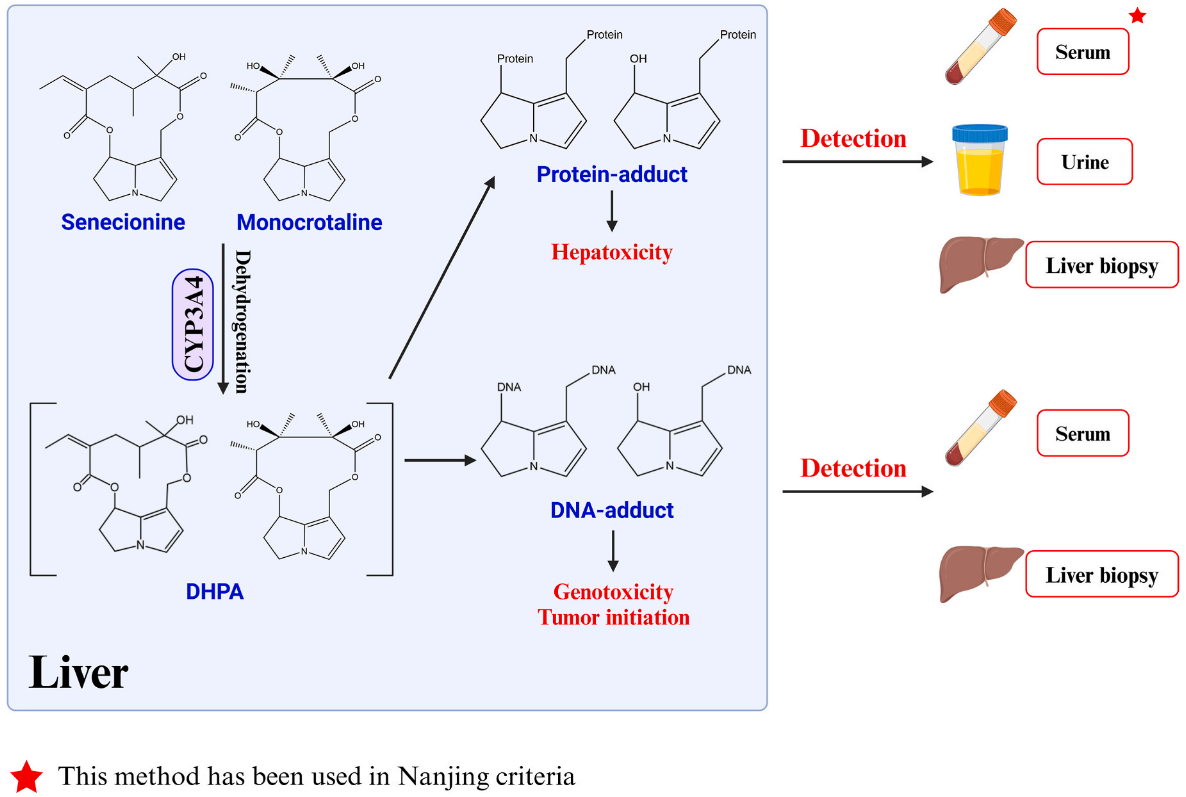


图1. 体内检测DHPA-DNA/蛋白质加合物的示意图。

4.1. 作为生物标志物的 miRNA

miRNA是一类内源性短小且高度保守的非编码RNA，由18–25个核苷酸组成，在包括发育、增殖、分化、凋亡和能量代谢等多种生理和病理过程中发挥关键作用（Bartel, 2004）。miRNA在血液中具有良好的稳定性，随着miRNA检测技术的发展，其检测变得更加容易，从而赋予miRNA作为新兴且微创的中药诱导毒性诊断生物标志物的巨大潜力（Bartel, 2004；Chen et al., 2008）。

先前的研究表明，给予泽泻（Dioscorea bulbifera L.）或其主要肝毒性化合物泽泻苷-B（diosbulbin-B）的老鼠血清中，miRNA-122-3p 和 miRNA-194-5p 显著上调。值得注意的是，这种增加发生在 ALT 或 AST 水平变化之前，ALT 和 AST 是常用于评估肝损伤的两个经典生化指标（Yang 等，2017b）。此外，给予泽泻或泽泻苷-B 的老鼠血清中 miRNA-5099 显著增加，而给予其他三种肝毒素（包括对乙酰氨基酚、单氯吡啶和 TSN）的老鼠中未观察到这种变化（ Yang 等，2017b）。除了发现 miRNA-122-3p 在给予 TSN、泽泻苷-B、对乙酰氨基酚和单氯吡啶的老鼠血清中水平上调之外，miRNA-367-3p 仅在接触 TSN 的老鼠中下降（Yang 等，2019）。先前的报告已证实 miRNA-122 在肝细胞中高度丰富，并具有作为诊断生物标志物的潜在应用价值。

在由马兜铃酸I（一种典型的AA）引起的肾毒性中，miRNA-21-3p在血浆和肾脏中显著升高，但在非靶器官中并未升高，而且这种增加与AA的剂量高度相关（Pu 等，2017）。值得注意的是，这种上调发生在传统肾损伤标志物如肌酐和血尿素氮（BUN）升高之前，以及肾小管损伤发生之前（Pu 等，2017）。这些发现表明，miRNA-21-3p有可能成为由AA及含AA的草药引起的肾毒性的敏感早期预警生物标志物。先前的研究表明，miRNA-21通常与药物性肾损伤（DIKI）相关，并且通过检测尿液中的miRNA-21可以预测DIKI的发生（Huang 等，2020）。由于检测尿液中的miRNA-21比血浆更容易，因为尿液检测是无创的，这使其成为临床监测AA诱导肾毒性的更实用的方法。

草酸，主要存在于杨桃（Averrhoa carambola L.）中，据报道可引起肾毒性（Shihana 等，2020）。一项先前的研究表明，与未受伤的患者相比，急性肾损伤患者（因摄入草酸中毒）体内的五种microRNA（miRNA-20a、miRNA-92a、miRNA-93、miRNA-195、miRNA-451）显著降低，并且与标准化尿白蛋白呈显著正相关，这表明这些miRNA有可能作为反映其肾毒性的生物标志物（Shihana 等，2020）。尽管这一发现具有潜力，但仍需进一步的实验研究和临床调查来验证这些miRNA在评估草酸或杨桃引起的肾毒性中的潜在应用价值。

雷公藤内酯是从雷公藤（Tripterygium wilfordii Hook. f.）中提取的一种有毒化合物，其多器官毒性已有充分文献记录，包括肝毒性（Kong 等，2015）、肾毒性（Sun 等，2013）和心脏毒性（Zhou 等，2015）。有研究显示，雷公藤内酯在大鼠中引起严重的心脏毒性，并表现为显著的

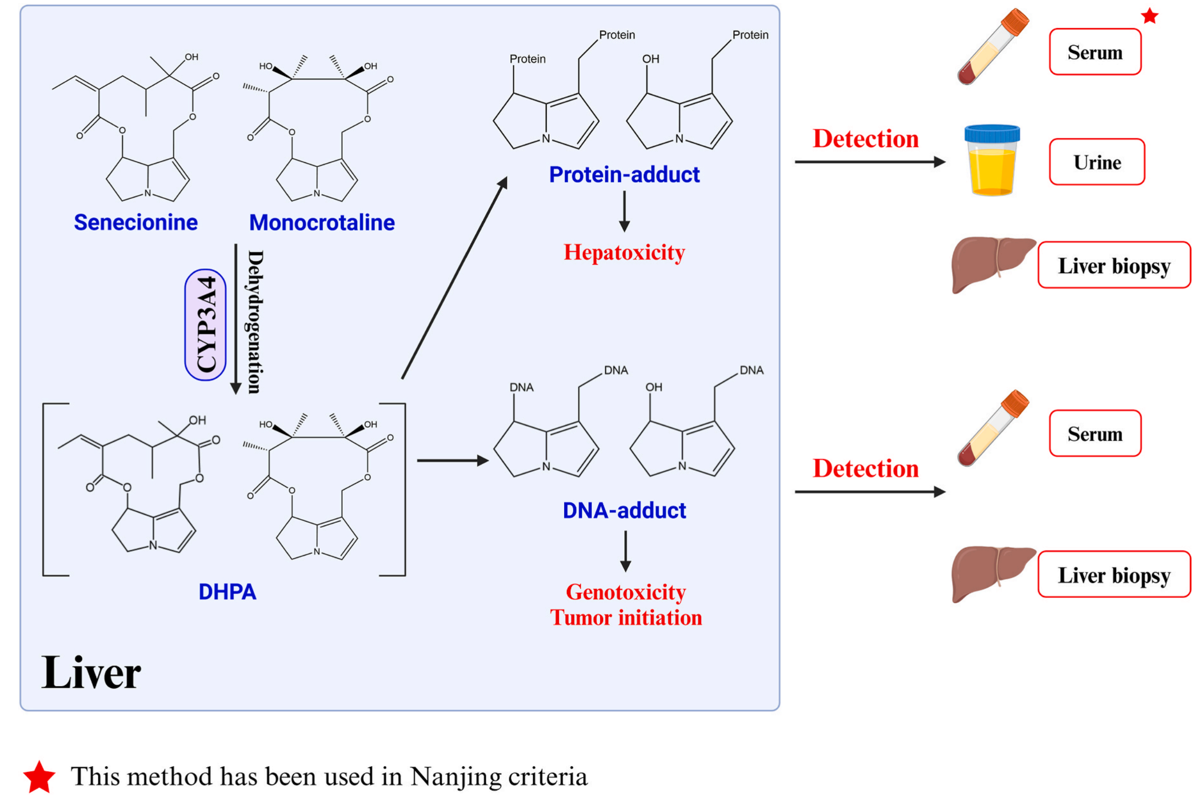


Fig. 1. Diagram of the detection of DHPA-DNA/protein adducts *in vivo*.

4.1. miRNA as the biomarker

miRNA is a class of endogenously short and highly conserved non-coding RNA composed of 18–25 nucleotides and displays a critical role in miscellaneous physiological and pathological processes including development, proliferation, differentiation, apoptosis, and energy metabolism (Bartel, 2004). miRNA has a good stability in the blood and can be easily detected with the development of miRNA detection technology, thereby endowing miRNA with the huge potential to be an emerging and minimally invasive biomarker for the diagnosis of TCM-induced toxicity (Bartel, 2004; Chen et al., 2008).

Previous research manifested that miRNA-122-3p and miRNA-194-5p were significantly up-regulated in the serum of mice given with *Dioscorea bulbifera* L. or its primary hepatotoxic compound diosbulbin-B. Notably, this increase occurred earlier than the change of ALT or AST, which are two classic biochemical indicators commonly used to assess liver injury (Yang et al., 2017b). Additionally, serum miRNA-5099 was significantly increased in mice treated with *Dioscorea bulbifera* L. or diosbulbin-B, but not in mice given with three other hepatotoxins, including acetaminophen, monocrotaline, and TSN (Yang et al., 2017b). Except finding that the level of miRNA-122-3p was up-regulated in serum from mice treated with TSN, diosbulbin-B, acetaminophen and monocrotaline, miRNA-367-3p was decreased only in mice exposed to TSN (Yang et al., 2019). Previous reports have already demonstrated the potential application of miRNA-122, which is highly abundant in hepatocytes, as a biomarker for the diagnosis of multifarious liver diseases such as patients with DILI (Su et al., 2012; Russo et al., 2017; Martin-Romero et al., 2020). However, the disadvantage of miRNA-122 is that only miRNA-122 could not distinguish the concrete type of liver disease. These above evidences further supports the usage of miRNA-122 as a universally sensitive biomarker for reflecting hepatic damage, but miRNA-5099 may serve as a specific biomarker for hepatotoxicity caused by *Dioscorea bulbifera* L., while miRNA-367-3p could be a specific biomarker for hepatotoxicity caused by TSN.

In nephrotoxicity induced by aristolochic acid I (a typical AA), miRNA-21-3p was markedly enhanced in both plasma and kidney, but not in non-target organs, and this increase was highly correlated with the dosage of AA (Pu et al., 2017). Notably, this upregulation occurred earlier than the elevation of traditional renal injury markers such as creatinine and BUN, as well as the onset of renal tubular injury (Pu et al., 2017). These finding suggest that miRNA-21-3p has the potential to be a sensitive early-warning biomarker for nephrotoxicity induced by AA and AA-containing herbal medicines. A previous study has manifested that miRNA-21 was generally associated with drug-induced kidney injury (DIKI), and the occurrence of DIKI can be predicted by detecting miRNA-21 in urine (Huang et al., 2020). As determining miRNA-21 in urine is easier than in plasma because urine-based detection is non-invasive, making it a more practical approach for clinical monitoring AA-induced nephrotoxicity. However, whether detecting miRNA-21 in urine can reliably predict nephrotoxicity specifically caused by AA needs further investigation.

Oxalic acid, primarily present in *Averrhoa carambola* L., has been reported to induce nephrotoxicity (Shihana et al., 2020). A previous study demonstrated that five microRNAs (miRNA-20a, miRNA-92a, miRNA-93, miRNA-195, miRNA-451) markedly decreased in acute renal damage patients intoxicated with oxalic acid as compared to patients with no injury and showed a markedly positive correlation with normalized urinary albumin, indicating the potential application of those miRNAs as biomarkers for reflecting its nephrotoxicity (Shihana et al., 2020). Despite this promising finding, further experimental studies and clinical investigations are needed to validate the potential application of these miRNAs for assessing nephrotoxicity induced by oxalic acid or *Averrhoa carambola* L.

Triptolide, a toxic compound from *Tripterygium wilfordii* Hook. f., is well-documented for its multiple organ toxicities, including hepatotoxicity (Kong et al., 2015), nephrotoxicity (Sun et al., 2013), and cardiotoxicity (Zhou et al., 2015). A study manifested that triptolide caused serious cardiotoxicity in rats, and characterized by significant miRNA表达谱的变化（Wang 等，2019）。在接受雷公藤内酯治疗的大鼠中，共有15种miRNA在心脏和血浆中上调。在这些miRNA中，miRNA-615和miRNA-127-5p在早期达到峰值，而miRNA-15a-3p和miRNA-483-3p的水平则逐渐升高，在晚期达到峰值（Wang 等，2019）。上述四种miRNA有望作为雷公藤内酯诱导心脏毒性的早期预警生物标志物，为其毒性机制提供有价值的见解，并为及时干预提供方法。

4.2. LncRNA 作为生物标志物

LncRNA 被定义为内源性非编码 RNA，其转录本长度超过 200 个核苷酸（Bridges 等, 2021），并且参与体内许多生物过程（Batista 和 Chang, 2013；Rinn 等, 2007）。一些先前的研究显示，lncRNA-ENST00000446135 和 lncRNA-ENST00000414355 可能是反映镉诱导毒性的新型生物标志物（Zhou 等, 2020；Moawad 等, 2021）。与此同时，一些研究指出 lncRNA 可能作为创新毒理学生物标志物，用于检测包括多柔比星等药物在内的多种化学物质诱导的毒性（Dempsey 和 Cui, 2017；Fa 等, 2021）。在雷公藤苷所致的肝损伤中，转录组学结果显示超过 400 种 lncRNA 的表达发生了显著变化，但这些 lncRNA 是否能够用作毒理学生物标志物仍需进一步实验验证（ An 等 , 2023 ）。目前，对于这一部分的研究仍然不多

表3 微小RNA（miRNA）作为中药引起的毒性生物标志物。

毒性类型	有毒的中医（化合物）	微小RNA	趋势	参考
肝毒性	DBL（地奥昔苷B）	miRNA-122-3页	↑	杨等人, 2017b
		miRNA-194-5页	↑	
		miRNA-5099	↑	
	MTSZ 毒杉酮	miRNA-122-3页	↑	杨等（2019）
		miRNA-367-3p	↓	
肾毒性	ARI（关木通类）酸 I	miRNA-21-3p	↑	蒲 等人（2017年）
	乙二酸（草酸）	miRNA-20a	↓	Shihana 等人（2020）
		miRNA-92a	↓	
		miRNA-93	↓	
		miRNA-195	↓	
		miRNA-451	↓	
心脏毒性	TWHF（雷公藤内酯）	miRNA-15a-3p	↑	王等（2019）
		miRNA-127-5p	↑	
		miRNA-483-3p	↑	
		miRNA-615	↑	

马兜铃属植物，*ARI*；杨桃，*AC*；薯蓣，*DBL*；苦楝，*MTSZ*；雷公藤，*TWHF*。

以上所有研究表明，miRNA 和 lncRNA 显示出成为中药诱导毒性诊断生物标志物的巨大潜力，因为它们在血清中易于测量，并且具有高灵敏度和特异性的优势。此外，以前的研究已经证明，一些 miRNA 显示出作为早期生物标志物反映中药诱导毒性的巨大潜力（表 3）。尽管关于 miRNA 作为毒性生物标志物的研究已经取得了一些进展，但仍然需要更多的临床数据和实验来将 miRNA 用作早期预警生物标志物。

5. 内源性代谢物作为中药诱导毒性的生物标志物

内源性代谢物是一类在全身代谢过程中产生的小分子化合物，它们对外界刺激作出反应，并且可以反映疾病的发生或直接参与其进展（Wishart, 2019；Qiu 等, 2023）。代谢组学作为一种强有力的分析技术，常用于分析和检测体液或组织样本中内源性代谢物的变化。凭借其快速分析、高灵敏度和高通量的特点，代谢组学已成为发现毒性新生物标志物的重要工具（Ni 等, 2014；Wishart, 2019）。它已经被广泛用于寻找中药的毒性生物标志物。

在接受非毒性剂量的脂多糖（LPS）与何首乌根（PMR，指何首乌干根块茎）及其主要成分2,3,5,4′-四羟基二苯乙烯（TSG）共同处理的大鼠血清中，共发现了32种内源性代谢物。在这些代谢物中，有十种生物标志物与TSG诱导的肝损伤相关，包括L-缬氨酸和L-脯氨酸水平降低，以及尿素、已酸等水平升高。值得注意的是，这些生物标志物中的六种还与PMR诱导的肝毒性相关（表4），提示这些血清代谢物可能作为PMR诱导肝损伤的潜在生物标志物（Lin 等，2021）。通过比较差异在从由PMR引起的肝损伤以及对PMR耐受的患者获得的血浆样本中，发现了8种潜在的生物标志物（表4）（Hu 等，2023）。在这些代谢物中，lysoPC (16:0)、次黄嘌呤和牛磺熊去氧胆酸在区分这两种情况方面显示出较高的敏感性和准确性，突显了其潜在的临床相关性（Hu 等，2023）。

已发现黑藜芦（Veratrum nigrum L.）可在小鼠中引起严重的肝损伤，随后采用超高效液相色谱-四极杆飞行时间质谱（UHPLC-Q-TOF-MS）分析鉴定出肝脏中的15种内源性代谢物（表4）。这些代谢物被提出作为黑藜芦所致肝损伤的潜在生物标志物（Wei 等，2017）。另一项在人类肝细胞中进行的代谢组学研究鉴定了8种代谢物，包括神经酰胺二醇、谷胱甘肽、L-棕榈酰肉碱、反式油酸肉碱等（表4），作为大黄素、雷公藤内酯和AA体外诱导的肝损伤的潜在生物标志物（Liu 等，2015）。与此同时，Liu 等报道指出，消痰中药五加皮（Dysosma Versipellis）可导致大鼠严重肝损伤。在他们的研究中，通过UHPLC-Q-TOF-MS分析初步鉴定出17种显著变化的化学化合物，包括5-羟基异枸橼酸、果糖甘氨酸等。这些发现表明，它们可能是潜在的生物标志物。

有趣的是，次黄嘌呤和溶血磷脂酰胆碱 (18:1) 在何首乌和乌头所致的肝毒性中均表现出显著变化，但它们的变化方向完全相反。同样，神经酰胺在乌头所致的肝毒性以及大黄素、雷公藤内酯和丙烯酰胺体外诱导的肝损伤中也发现有变化，而溶血磷脂酰胆碱 (20:3) 则报道在乌头和半边莲所致的肝毒性中发生变化。然而，在这些情况下，它们的变化趋势也完全相反。次黄嘌呤是嘌呤代谢的中间产物，目前尚未被报道为毒性潜在生物标志物。另一方面，神经酰胺作为鞘脂合成的中间代谢物，已经被确定为潜在的

alternations in miRNA expression profiles (Wang et al., 2019). A total of 15 miRNAs were up-regulated in heart and plasma in rats treated with triptolide. Among those miRNAs, miRNA-615 and miRNA-127-5p reached the peak at the early stage, while the level of miRNA-15a-3p and miRNA-483-3p increased gradually and reached the peak at the late stage (Wang et al., 2019). These above four miRNAs hold potential as early-warning biomarkers for triptolide-induced cardiotoxicity, providing valuable insights into its toxic mechanism and offering the approach for timely intervention.

4.2. LncRNA as the biomarker

LncRNA is defined as endogenous non-coding RNA whose transcript is more than 200 nucleotides (Bridges et al., 2021), and it participates in many biological processes in the body (Batista and Chang, 2013; Rinn et al., 2007). Some previous studies displayed that lncRNA-ENST00000446135 and lncRNA-ENST00000414355 might be novel biomarkers for reflecting cadmium-induced toxicity (Zhou et al., 2020; Moawad et al., 2021). Meanwhile, some researches pointed out the possibility of lncRNA as innovative toxicological biomarkers for toxicity induced by a variety of chemicals including some drugs like doxorubicin (Dempsey and Cui, 2017; Fa et al., 2021). In the liver damage induced by tripterygium glycosides, results from transcriptomics have revealed significant alternations in the expression of over 400 lncRNAs, but whether these lncRNAs can be used as toxicological biomarkers requires further experimental validation (An et al., 2023). Currently, there is still no much research about the participation of lncRNA in toxicity induced by TCM. More attentions should be directed towards uncovering the potential application of lncRNAs as biomarkers for toxicity induced by TCM, as this represents a promising yet underexplored area of research.

All above researches have illustrated that miRNA and lncRNA showed a huge potential to be diagnostic biomarkers for toxicity induced by TCM, because they are readily measurable in serum and with the advantage of high sensitivity and specificity. Moreover, previous studies have already evidenced that some miRNAs showed the huge potential to be used as early biomarkers to reflect toxicity induced by TCM (Table 3). Although the study on miRNA as biomarkers for toxicity has made some progressions, it is still of necessity that more clinical data and experiments are needed for using miRNAs as early-warning biomarkers.

5. Endogenous metabolites as biomarkers for toxicity induced by TCM

Endogenous metabolites are a collection of small molecular compounds produced during metabolic processes in the whole body in response to external stimuli, and they can reflect the occurrence of disease or directly participate in its progression (Wishart, 2019; Qiu et al., 2023). Metabolomics, as a powerful analytical technique, is often used to analyze and detect the change of endogenous metabolites in body fluids or tissue samples. With its rapid analysis, high sensitivity and high throughput, metabolomics has become a valuable tool for discovering new biomarkers for toxicity (Ni et al., 2014; Wishart, 2019). It has already been widely used to find the toxic biomarkers of TCM.

A total of 32 types of endogenous metabolites were found in the serum from rats co-treated with a non-toxic dose of LPS and Polygonum multiflorum radix (PMR), which is the dried root tuber of *Polygonum multiflorum* Thunb., as well as its main compound 2,3,5,4′-tetrahydroxyldiphenylethylene (TSG). Among these metabolites, ten biomarkers were associated with TSG-induced hepatic damage, including reduced levels of L-valine and L-proline, and increased levels of urea, caproic acid, among others. Notably, six of these biomarkers were also found to be related to the PMR-induced hepatotoxicity (Table 4), suggesting that these serum metabolites could serve as potential biomarkers for PMR-induced liver injury (Lin et al., 2021). By comparing the differential

Table 3
MicroRNA (miRNA) as the biomarker for toxicity induced by TCM.

Type of toxicity	Toxic TCM (compounds)	miRNA	Trend	Reference
Hepatotoxicity	DBL (Diosbulbin B)	miRNA-122-3p	↑	Yang et al., 2017b
		miRNA-194-5p	↑	
		miRNA-5099	↑	
	MTSZ (Toosendanin)	miRNA-122-3p	↑	Yang et al. (2019)
		miRNA-367-3p	↓	
Nephrotoxicity	ARI (Aristolochic acid I)	miRNA-21-3p	↑	Pu et al. (2017)
	AC (Oxalic acid)	miRNA-20a	↓	Shihana et al. (2020)
		miRNA-92a	↓	
		miRNA-93	↓	
		miRNA-195	↓	
		miRNA-451	↓	
Cardiotoxicity	TWHF (Triptolide)	miRNA-15a-3p	↑	Wang et al. (2019)
		miRNA-127-5p	↑	
		miRNA-483-3p	↑	
		miRNA-615	↑	

Aristolochia species, ARI; *Averrhoa carambola*, AC; *Dioscorea bulbifera* L., DBL; *Melia toosendan Sieb.et Zucc.*, MTSZ; *Tripterygium wilfordii* Hook.f., TWHF.

metabolites in plasma samples obtained from PMR-induced liver injury and PMR-tolerance patients, 8 potential biomarkers were discovered (Table 4) (Hu et al., 2023). Among these metabolites, lysoPC (16:0), hypoxanthine, and taurochenodesoxycholic acid demonstrated the high sensitivity and accuracy in distinguishing between these two conditions, highlighting their potential clinical relevance (Hu et al., 2023).

Veratrum nigrum L. has been found to induce severe hepatic injury in mice, and subsequent analysis using UHPLC-Q-TOF-MS identified 15 endogenous metabolites in the liver (Table 4). These metabolites were proposed as potential biomarkers for hepatic injury caused by *Veratrum nigrum* L. (Wei et al., 2017). Another metabolomics study conducted in human hepatocytes identified 8 metabolites, including sphinganine, glutathione, l-palmitoylcarnitine, elaidic carnitine and so on (Table 4), as potential biomarkers for hepatic injury induced by emodin, triptolide, and AA *in vitro* (Liu et al., 2015). Meanwhile, Liu et al. reported that *Dysosma Versipellis*, a traditional Chinese medicine for eliminating phlegm, could cause severe liver damage in rats. In their study, 17 chemical compounds that significantly changed were preliminarily identified based on UHPLC-Q-TOF-MS analysis, including 5-hydroxyisourate, fructoseglycine, and so on. These findings suggest that they could be potential biomarkers for toxicity induced by *Dysosma Versipellis* (Liu et al., 2020).

Interestingly, hypoxanthine and lysoPC (18:1) exhibited significant changes in hepatotoxicity induced by both *Polygonum multiflorum* Thunb. and *Veratrum nigrum* L., but they showed the completely opposite direction. Similarly, sphinganine was found to be changed in hepatotoxicity induced by *Veratrum nigrum* L. and hepatic injury induced by emodin, triptolide, and AA *in vitro*, and lysoPC (20:3) was reported to be changed in hepatotoxicity induced by both *Veratrum nigrum* L. and *Dysosma Versipellis*. However, their trend of change was also completely opposite in these cases. Hypoxanthine, an intermediate product of purine metabolism, has not yet been reported as a potential biomarker for toxicity. On the other hand, sphinganine, an intermediate metabolite of sphingolipid synthesis, has already been identified as a potential

表4 中药诱导毒性相关的内源性代谢物作为生物标志物。

毒性类型	有毒中药（复方）	代谢物	趋势	参考
肝毒性	**PMR**	胆碱	↓	胡等人（2023）
		脱氧胆酸	↓	
		D-谷氨酸	↓	
		次黄嘌呤	↑	
		溶血磷脂酰胆碱 (16:0)	↑	
		溶血磷脂酰胆碱 (18:1)	↓	
		苯丙氨酰苯丙氨酸	↓	
		牛磺脱氧胆酸	↑	
	TSG	3-羟基丁酸	↑	林等人（2021年）
		己酸	↑	
		D-半乳糖	↑	
		DL-苹果酸	↑	
		D-甘露糖	↑	
		十六酸	↑	
		L-脯氨酸	↓	
		L-缬氨酸	↓	
		十八烷	↑	
		Urea	↑	
	VNL	十六烯酸	↑	魏等人（2017年）
		次黄嘌呤	↓	
		次黄嘌呤核苷	↑	
		溶血磷脂酰胆碱 (16:1)	↑	
		溶血磷脂酰胆碱 (18:1)	↑	
		溶血磷脂酰胆碱 (20:2)	↑	
		溶血磷脂酰胆碱 (20:3)	↑	
		十八碳三烯酸	↑	
		十八碳二烯酸	↑	
		棕榈酰肉碱	↑	
		神经氨醇	↓	
		牛磺酸	↓	
		色氨酸	↑	
		尿酸	↑	
		尿苷	↑	
	DV	5-羟基异尿酸	↑	刘等人（2020）
		果糖甘氨酸	↓	
		十三酰甘氨酸	↓	
		DHAP (8:0)	↓	
		12-二十三醇	↑	
		硬脂酰甘氨酸	↓	
		dUDP	↓	
		花生酰肉碱	↓	
		dUTP	↓	
		溶血磷脂酰胆碱 (20:3)	↓	
		L-肉碱	↓	
		苯丙氨酸	↑	
		D-核糖-5-磷酸	↑	
		雌二醇-17β-3-硫酸酯	↑	
		溶血磷脂酰胆碱 (18:0)	↑	
		色氨酸-脯氨酸	↓	
		组氨酰-苯丙氨酸	↓	
	VMG（大黄素）-ARI（关木通酸）-TWHF（雷公藤内酯）	C16 鞘氨醇	↑	刘等人（2015）
		反式肉碱	↑	
		谷氨酸 甘氨酸	↑	
		谷胱甘肽	↓	
		L-棕榈酰肉碱	↑	
		N1-乙酰基亚精胺	↑	
		N-十一酰基甘氨酸	↓	
		神经氨醇	↑	
肾毒性	**SNL**	19-羟基脱氧皮质酮	↑	罗等人（2018）
		L-肉碱	↓	
		LPE (18:2)	↑	
	TWHF	3-吲哚丙酸	↓	
		LPE (20:1)	↓	
		溶血磷脂酰胆碱 (18:2)	↑	
		溶血磷脂酰胆碱 (20:5)	↓	
		溶血磷脂酰胆碱 (22:5)	↓	
	SNL 和 TWHF	丙氨酸	↓	
		胆酸	↓	
		异丁酰-L-肉碱	↓	
		溶血磷脂酰胆碱 (14:0)	↓	
		溶血磷脂酰胆碱 (16:1)	↓	
		溶血磷脂酰胆碱 (18:4)	↓	
		溶血磷脂酰胆碱 (20:2)	↓	
		溶血磷脂酰胆碱 (20:3)	↓	

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Table 4
Endogenous metabolites as biomarkers for toxicity induced by TCM.

Type of toxicity	Toxic TCM (Compound)	Metabolite	Trend	Reference
Hepatotoxicity	**PMR**	Choline	↓	Hu et al. (2023)
		Deoxycholic acid	↓	
		D-Glutamic acid	↓	
		Hypoxanthine	↑	
		LysoPC (16:0)	↑	
		LysoPC (18:1)	↓	
		Phenylalanylphenylalanine	↓	
		Taurochenodesoxycholic acid	↑	
	TSG	3-hydroxybutyric acid	↑	Lin et al. (2021)
		Caproic acid	↑	
		D-galactose	↑	
		DL-malic acid	↑	
		D-mannose	↑	
		Hexadecanoic acid	↑	
		L-proline	↓	
		L-valine	↓	
		Octadecane	↑	
		Urea	↑	
	VNL	Hexadecenoic acid	↑	Wei et al. (2017)
		Hypoxanthine	↓	
		Inosine	↑	
		LysoPC (16:1)	↑	
		LysoPC (18:1)	↑	
		LysoPC (20:2)	↑	
		LysoPC (20:3)	↑	
		Octadecatrienoic acid	↑	
		Octadecadienoic acid	↑	
		Palmitoylcarnitine	↑	
		Sphinganine	↓	
		Taurine	↓	
		Tryptophan	↑	
		Uric acid	↑	
		Uridine	↑	
	DV	5-Hydroxyisourate	↑	Liu et al. (2020)
		Fructoseglycine	↓	
		Tridecanoylglycine	↓	
		DHAP (8:0)	↓	
		12-Tricosanol	↑	
		Stearoylglycine	↓	
		dUDP	↓	
		Arachidylcarnitine	↓	
		dUTP	↓	
		LysoPC (20:3)	↓	
		L-carnitine	↓	
		Phenylalanine	↑	
		D-Ribose5-Phosphate	↑	
		Estradiol-17beta3-sulfate	↑	
		LysoPC (18:0)	↑	
		Tryptophyl-Proline	↓	
		Histidinyl-Phenylalanine	↓	
	VMG (Emodin) -ARI (Aristolochic acid)-TWHF (Triptolide)	C16 sphinganine	↑	Liu et al. (2015)
		Elaidic carnitine	↑	
		Glu Gly	↑	
		Glutathione	↓	
		L-Palmitoylcarnitine	↑	
		N1-Acetylspermidine	↑	
		N-Undecanoylglycine	↓	
		Sphinganine	↑	
Nephrotoxicity	**SNL**	19-hydroxydeoxycorticosterone	↑	Luo et al. (2018)
		L-carnitine	↓	
		LPE (18:2)	↑	
	TWHF	3-indolepropionic acid	↓	
		LPE (20:1)	↓	
		LysoPC (18:2)	↑	
		LysoPC (20:5)	↓	
		LysoPC (22:5)	↓	
	SNL and TWHF	Alanine	↓	
		Cholic acid	↓	
		Isobutyryl-L-carnitine	↓	
		LysoPC (14:0)	↓	
		LysoPC (16:1)	↓	
		LysoPC (18:4)	↓	
		LysoPC (20:2)	↓	
		LysoPC (20:3)	↓	

(*continued on next page*)

表 4（续）

毒性类型	有毒中药（复方）	代谢物	趋势	参考
		棕榈酰肉碱	↓	
		苯丙氨酸	↓	
		胸腺嘧啶核苷	↓	
		色氨酸	↓	

马兜铃属植物，ARI；淫羊藿，DV；何首乌根，PMR。马钱子，SNL；2,3,5,4′-四羟基二苯乙烯，TSG；雷公藤，TWHF；印度黄藤，VMG；黑藜芦，VNL。

在长期暴露于伏马毒素的鸭子中，毒性生物标志物（Tran 等，2006）。LysoPC (18:1) 和 LysoPC (20:3) 都归类为溶血磷脂酰胆碱，是在多种生物过程中起多样作用的关键磷脂。已有研究发现，lysoPC 水平的变化与肝损伤相关（Shi 等，2016；Furuhashi，2020）。虽然其在不同肝损伤中的变化趋势有所不同，但观察到的次黄嘌呤、鞘氨醇、lysoPC (18:1) 和 lysoPC (20:3) 的变化表明，它们可能作为早期检测和评估肝毒性的潜在生物标志物。

在由槟榔（Strychnos nux-vomica Linn.）或雷公藤（Tripterygium wilfordii Hook F.）引起的肾毒性中，血浆代谢组分析结果鉴定出与槟榔相关的15种代谢物和与雷公藤相关的17种代谢物（表 4）（Luo 等，2018）。在这些鉴定的代谢物中，有12种在由这两种草药引起的肾毒性中表现出共同的改变。这些共享的代谢物包括丙氨酸、胆酸等，凸显了它们在肾毒性病理过程中的关键作用。这些结果表明，这些代谢物作为早期检测和监测由槟榔或雷公藤引起的肾毒性的潜在生物标志物具有巨大潜力。

代谢组学在内源性代谢物的定性和定量分析方面具有显著优势，但由于现有数据库的限制和检测灵敏度的降低，其识别潜在毒性生物标志物的能力仍然有限。尽管代谢组学为发现潜在生物标志物提供了有价值的见解和初步线索，但仍需通过广泛的临床验证来确认其在临床实践中的适用性和可靠性。通过改进方法学和建立健全的数据库来解决这些局限性，将有助于增强代谢组学在发现中药引起的毒性生物标志物方面的应用价值。

6. 内源性蛋白作为中药诱导毒性的生物标志物

一些蛋白质因与机体的生理功能和病理状态密切相关而成为有价值的毒性生物标志物。这些蛋白质的表达、浓度或活性的变化能够敏感且特异地反映药物的毒性作用，为其对机体的影响提供关键性见解。因此，这些蛋白质作为中药诱导的毒性检测、监测和评估的生物标志物具有重要潜力。

细胞因子是由多种细胞分泌的可溶性蛋白，具有多种生物功能。由于其高易感性和易检测性，将细胞因子视为早期预警生物标志物的候选者是合理的（Tu 等，2021）。有报道表明，轻度免疫应激可能是由何首乌引起的特异性药物性肝损伤（idiosyncratic DILI）易感性相关的因素之一（Boelsterli 和 Lee，2014；Tu 等，2019；Rao 等，2021）。先前的一项研究显示，12 种与炎症和免疫调节相关的趋化因子 [干扰素-γ诱导的蛋白 10、单核细胞趋化蛋白-1（MCP-1）等] 可被认为是识别何首乌引起肝毒性易感性的潜在生物标志物（Tu 等，2019）。最近，两项独立的临床回顾性研究结果显示，包括肿瘤坏死因子 α（TNFα）、MCP-1 和血管内皮生长因子在内的细胞因子也与此相关。

与耐受何首乌人群相比，何首乌诱导的药物性肝损伤（DILI）患者显示出差异。这些结果表明，TNFα、MCP-1 和 VEGF 可能作为预测对何首乌相关 DILI 易感性的可靠生物标志物（Tu 等，2021）。先前的研究显示，小鼠血清中 CD30 配体和白细胞介素-3 的表达变化可能是由山药薯蓣引起的肝损伤的潜在生物标志物（Sheng 等，2014）。这些发现共同强调了细胞因子在预测和诊断中药诱导肝毒性方面的潜力。

肾损伤分子-1（KIM-1）是一种I型细胞膜糖蛋白。先前的研究表明，在正常情况下，KIM-1的表达几乎无法检测到，但在肾损伤后，肾细胞中的KIM-1显著上调（Ichimura 等，1998, 2008；Griffin 等，2019）。目前，KIM-1已成为急性肾损伤的新型且敏感的生物标志物，并已被用于非临床和临床药物开发中，以帮助美国食品药品监督管理局和欧洲药品管理局进行安全性评估（Bonventre 等，2010；Dieterle 等，2010；Abdel-salam 等，2018）。在小鼠AA诱导的肾毒性中，尿液中KIM-1的升高比传统肾损伤指标更早，并且仅需25 μ微升尿液即可检测到KIM-1，其稳定性至少可达5天（Sabbisetti 等，2013）。这些结果表明，KIM-1也可能作为识别AA诱导肾毒性的有价值生物标志物。通过使用

在先前的一项研究中，采用了急性毒性、累积毒性、耐受性测试和亚慢性毒性研究等多种方法来评估山豆根的肝毒性。研究结果显示，血清胆碱酯酶（CHE）水平呈剂量依赖性升高，而在亚慢性毒性研究中，ALT和AST等传统肝损伤生物标志物呈下降趋势。这表明血清CHE的升高可能作为评估山豆根诱导肝损伤的潜在辅助生物标志物（Wang 等，2017）。在一项关于栀子苷诱导肝毒性的研究中，发现甘氨酸 N-甲基转移酶和糖原磷酸化酶两种生物标志物能够比经典生物标志物更早显著地提示肝损伤（Wei 等，2014）。然而，还需要进一步的实验研究和临床证据来确保其可靠性。

在菝葜茎中发现的皂苷多叶甙，包括多叶甙VI和多叶甙VII，据报道可通过干扰胆固醇生物合成引起肝毒性，导致肝细胞凋亡和脂质代谢紊乱（Li等，2022）。已发现角鲨烯合酶（SQLE）在调节胆固醇生物合成中起核心作用，而另一种多叶甙——多叶甙I被发现可直接与SQLE结合。因此，SQLE被认为是评估菝葜茎诱导肝损伤的潜在生物标志物（Li

Table 4 (*continued*)

Type of toxicity	Toxic TCM (Compound)	Metabolite	Trend	Reference
		Palmitoylcarnitine	↓	
		Phenylalanine	↓	
		Thymidine	↓	
		Tryptophan	↓	

Aristolochia species, ARI; *Dysosma Versipellis*, DV; Polygonum multiflorum Radix, PMR. *Strychnos nux-vomica* Linn., SNL; 2,3,5,4′-tetrahydroxyl-diphenylethylene, TSG; *Tripterygium wilfordii* Hook F., TWHF; *Ventilago maderaspatana* Gaertn.; VMG; *Veratrum nigrum* L., VNL.

biomarker of toxicity in ducks chronically exposed to fumonisin (Tran et al., 2006). LysoPC (18:1) and LysoPC (20:3), both classified as lysophosphatidylcholines, are key phospholipids that play diverse roles in a variety of biological processes. Alternations in lysoPC levels have already been found to be associated with liver injury (Shi et al., 2016; Furuhashi, 2020). Although their trends of change vary in different liver injuries, the observed alteration in hypoxanthine, sphinganine, lysoPC (18:1), and lysoPC (20:3) suggest that they may serve as potential biomarkers for the early detection and evaluation of hepatotoxicity.

In nephrotoxicity induced by *Strychnos nux-vomica* Linn. or *Tripterygium wilfordii* Hook F., plasma metabolic profiling analysis results identified 15 metabolites associated with *Strychnos nux-vomica* Linn. and 17 metabolites related to *Tripterygium wilfordii* Hook F. (Table 4) (Luo et al., 2018). Among these identified metabolites, 12 were found to be commonly altered in nephrotoxicity caused by both herbal medicines. These shared metabolites include alanine, cholic acid, and others, highlighting their critical role in the pathological processes of nephrotoxicity. These results suggest the huge potential of these metabolites as promising biomarkers for early detection and monitoring nephrotoxicity induced by *Strychnos nux-vomica* Linn. or *Tripterygium wilfordii* Hook F.

Metabolomics exhibits significant advantages in the qualitative and quantitative analysis of endogenous metabolites, but its ability to identify potential biomarkers for toxicity remains limited due to constraints in existing databases and reduced sensitivity in detection. Although metabolomics offers valuable insights and initial clues for discovering potential biomarkers, extensive clinical validations are required to confirm their applicability and reliability in clinical practice. Addressing these limitations through improved methodologies and robust databases will be helpful for enhancing the utility of metabolomics in discovering biomarkers for toxicity induced by TCM.

6. Endogenous proteins as biomarkers for toxicity induced by TCM

Some proteins serve as valuable biomarkers for toxicity because they are closely associated with the physiological functions and pathological states of the body. Changes in the expression, concentration, or activity of these proteins can sensitively and specifically reflect the toxic effects of drugs, providing critical insights into their impact on the body. Hence, these proteins hold significant potential as biomarkers for detecting, monitoring, and assessing toxicity induced by TCM.

Cytokines are soluble proteins secreted from a variety of cells with multiple biological functions. It is reasonable for considering cytokines as promising candidates for early-warning biomarkers due to their high susceptibility and ease of detection (Tu et al., 2021). It has been reported that mild immune stress could be one of the susceptibility-related factors of idiosyncratic DILI caused by Polygoni multiflori Radix (Boelsterli and Lee, 2014; Tu et al., 2019; Rao et al., 2021). A previous study revealed that 12 chemokines [interferon-γ-inducible protein 10, monocyte chemotactic protein-1 (MCP-1), etc.] related to inflammation and immune regulation could be considered as potential biomarkers for identifying the susceptibility to hepatotoxicity caused by Polygoni multiflori Radix (Tu et al., 2019). Recently, results from two independent retrospective studies in clinic revealed that cytokines including tumor necrosis factor alpha (TNFα), MCP-1 and vascular endothelial growth factor (VEGF) markedly increased in the serum from patients with Polygoni multiflori Radix-induced DILI compared to Polygoni multiflori Radix-tolerant individuals. These results indicate that TNFα, MCP-1 and VEGF could serve as reliable biomarkers for predicting susceptibility to Polygoni multiflori Radix-related DILI (Tu et al., 2021). A previous study manifested that the altered expression of cluster of differentiation 30 ligand and interleukin-3 in the serum of mice could be potential biomarkers for liver injury caused by *Dioscorea bulbifera* L. (Sheng et al., 2014). These findings collectively underscore the potential of cytokines in predicting and diagnosing TCM-induced hepatotoxicity.

Kidney injury molecule-1 (KIM-1) is a type I cell membrane glycoprotein. Previous reports have demonstrated that KIM-1's expression was nearly undetectable under normal conditions but is markedly upregulated in renal cells following kidney damage (Ichimura et al., 1998, 2008; Griffin et al., 2019). Currently, KIM-1 has already been a novel and sensitive biomarker for acute kidney damage and been used as a biomarker in non-clinical and clinical drug development to help guide safety assessment by the Food and Drug Administration and European Medicines Agency (Bonventre et al., 2010; Dieterle et al., 2010; Abdel-salam et al., 2018). The increase of urine KIM-1 was earlier than classical indicators for kidney injury in AA-caused nephrotoxicity in mice, and KIM-1 could be detected with only 25 μl of urine and was stable for at least 5 days (Sabbisetti et al., 2013). These results imply that KIM-1 may also serve as a valuable biomarker for identifying nephrotoxicity induced by AA. By using KIM-1 as a biomarker for renal injury, some previous studies demonstrated that esculentoside A from the root of *Phytolacca acinose* Roxb. could cause significant nephrotoxicity (Gu et al., 2024). Similarly, although the methanol extract of *Tetrorchidium didymostemon* leaves had no obvious effect on body weight, biochemical parameters, oxidative injury indices and histopathology, the increased KIM-1 level alerted people to its potential nephrotoxicity (Ebohon et al., 2020). These observations underscore the potential of KIM-1 as a promising biomarker for detecting and assessing nephrotoxicity induced by TCM or its toxic compounds.

In a previous study, various methods such as acute toxicity, cumulative toxicity, tolerance testing, and sub-chronic toxicity studies were adopted to evaluate the liver toxicity of *Sophorae tonkinensis* Radix et Rhizoma. The findings revealed that the level of serum cholinesterase (CHE) increased in a dose-dependent manner, while traditional liver injury biomarkers such as ALT and AST showed a downward trend in the sub-chronic toxicity study. This suggests that the elevation of serum CHE may serve as a potential auxiliary biomarker for evaluating liver injury induced by *Sophorae tonkinensis* Radix et Rhizoma (Wang et al., 2017). In a study of hepatotoxicity induced by Geniposide from Gardeniae Fructus, two biomarkers including glycine N-methyltransferase and glycogen phosphorylase were found to indicate hepatic damage significantly earlier than classic biomarkers (Wei et al., 2014). However, further experimental investigations and clinical evidences are required to ensure their reliability and accuracy in assessing hepatotoxicity.

Polyphyllins found in Paridis Rhizoma, including polyphyllin VI and polyphyllin VII, have been reported to induce hepatotoxicity by disrupting cholesterol biosynthesis, leading to hepatocyte apoptosis and lipid metabolism disorders (Li et al., 2022). Squalene synthase (SQLE) has been found to play a central role in regulating cholesterol biosynthesis, and polyphyllin I (another polyphyllin) was discovered to directly bind with SQLE. Consequently, SQLE is considered a potential biomarker for assessing liver injury induced by Paridis Rhizoma (Li et al., 2023)。然而，尽管有这些发现，关于乌头参根引起的肝损伤的临床证据仍然有限，而且SQLE作为可靠生物标志物的作用尚未得到充分验证（Ding et al., 2021）。需要进一步深入的研究来确认SQLE在肝损伤中的临床相关性，并阐明多叶皂苷肝毒性作用的机制。

Diosbulbin D 是另一种来源于薯蓣（Dioscorea bulbifera L.）的有毒化合物，据报道它会通过细胞色素 P450 酶介导的肝脏代谢。这一代谢过程产生一种活性代谢物，能够共价结合微粒体环氧水解酶，从而触发抗微粒体环氧水解酶自身抗体的形成（Chen 等，2011）。值得注意的是，以前的研究表明，这些自身抗体可以在血清中长期存在（De Berardinis 等，2000）。鉴于它们在血清中的长期存在，抗微粒体环氧水解酶自身抗体有可能成为评估 Diosbulbin D 肝毒性的有前景的生物标志物。然而，其临床应用需通过全面的实验研究和可靠的临床证据进一步验证。

研究表明，其他蛋白质如β2-微球蛋白、脂钙蛋白2、骨桥蛋白和半胱氨酸蛋白酶抑制因子C，也可能作为传统草药引起的肾毒性的潜在生物标志物（Gao 等, 2023; Fuchs 等, 2014）。上述蛋白已被确定为评估中药及其毒性化合物引起毒性的潜在生物标志物（表5）。然而，还需要更多的临床证据来验证它们是否能够真正作为中药引起的毒性的生物标志物。

表5 中医药诱导毒性相关的蛋白质作为生物标志物。

Type	蛋白质	趋势	参考
细胞因子	白介素-1β	↑	Tu 等（2019）；Tu 等（2021）
	白细胞介素-6	↑	
	白细胞介素-10	↑	
	干扰素γ	↑	
	生长相关癌基因阿尔法	↑	
	粒细胞-巨噬细胞集落刺激因子	↑	
	肿瘤坏死因子α	↑	
	单核细胞趋化蛋白质-1	↑	
	单核细胞趋化蛋白质-3	↑	
	巨噬细胞炎症蛋白质-1 α	↑	
	激活后受调控正常T细胞表达并分泌	↑	
	干扰素γ诱导的蛋白质-10	↑	
	血管内皮生长因素	↑	
	分化群30配体	↑	盛等人（2014）
	白细胞介素-3	↓	
其他蛋白质	肾损伤分子-1	↑	Sabbisetti 等人 (2013); Gao 等, 2023; Ebohon 等人（2020）
	糖原磷酸化酶	↑	魏等人（2014年）
	甘氨酸N-甲基转移酶	↑	
	胆碱酯酶	↑	王等人（2017）
	角鲨烯合酶	↑	李等人（2023）
	抗微粒体环氧化物水解酶自体抗体	↑	陈等人（2011年）
	β2-微球蛋白	↑	顾等（2024）；傅赫斯等人（2014）
	脂质运载蛋白2	↑	
	骨桥蛋白	↑	
	胱抑素C	↑	

7. 结论

在本文中，我们系统地讨论并总结了中药诱导毒性的生物标志物，并以逻辑且全面的方式对其进行了分类。这些生物标志物具有高灵敏度和特异性，使其适合在临床中用于反映中药引起的毒性损伤。

首先，我们探讨了DNA/蛋白质加合物作为中药（TCM）诱导毒性生物标志物的显著潜力。这些加合物显示出高度的损伤特异性，能够准确判断中药诱导的损伤类型。例如，吡咯-蛋白加合物可指示含有PAs的中药引起的肝损伤（Lin 等，2011），而马兜铃酸-DNA加合物则可反映含有AA的中药引起的肾损伤（Jelakovic 等，2012）。然而，将DNA/蛋白质加合物用作生物标志物仍存在一些不足，因为其检测需要如LC/MS等精密仪器，从而导致检测成本较高。此外，测量过程还易受各种因素影响，如用药时间及其他类型加合物的潜在干扰。此外，加合物与中药诱导毒性之间的相关性还需要通过大规模临床试验进一步验证，以确保其可靠性和适用性。

非编码RNA由于其高敏感性和组织特异性，已成为中药毒性潜在的早期预警生物标志物。例如，有报道指出，含有巴戟天苷B的中药引起肝损伤时，miRNA-122-3p会迅速升高，其升高发生在ALT和AST等经典生物标志物变化之前。同样，血清miRNA-5099在实验中被发现是巴戟天苷B引起损伤的特异性生物标志物（Yang等，2017b）。同时，循环非编码RNA可以通过多种RNA测序方法检测，使用起来更加方便。然而，目前对非编码RNA的生物功能和分子机制的理解仍然有限。因此，有必要阐明非编码RNA差异表达在中药毒性反应中的重要性。同时也不可忽视的是，需要开发更有效、准确且可行的RNA检测方法以满足要求。

代谢物作为生物标志物具有明显优势，因为它们能够反映组织或器官的病理变化，而疾病的代谢特征可以用于评估风险或提供特定的疾病亚型。然而，在将代谢物作为中药诱导毒性的新型生物标志物应用时仍存在重大挑战。主要问题在于代谢物的识别和验证困难，这在很大程度上是由于当前代谢组学技术的局限性，代谢组学的覆盖范围仍然有限。另一方面，代谢变化非常复杂，目前尚无统一的策略或全面的分析平台来分析整个表型的所有代谢物。为了应对这些限制，建立统一、标准化和普遍认可的协议是必不可少的。考虑到高通量分析、数据整合以及个体差异的复杂性和难度...

其他潜在的生物标志物，如细胞因子、KIM-1 等，也被认为相较于经典生物标志物具有较高的敏感性，为进一步开发作为评估中药引起毒性标志物提供了有希望的前景。然而，上述审查的生物标志物仍然存在自身的不足。显然，新的早期预警生物标志物的识别和验证需要更广泛的实验研究和可靠的临床数据。展望未来，开发适当的联合诊断策略并整合多种生物标志物进行联合诊断可能是主要的发展方向。例如，高特异性的 DNA/蛋白加合物可用于确定损伤类型。

et al., 2023). However, despite these findings, clinical evidences regarding liver injury caused by Paridis Rhizoma remains limited, and the role of SQLE as a reliable biomarker has yet to be fully validated (Ding et al., 2021). Further in-depth studies are needed to confirm the clinical relevance of SQLE in liver injury and to elucidate the mechanisms underlying the hepatotoxic effects of polyphyllins.

Diosbulbin D, another toxic compound derived from *Dioscorea bulbifera* L., has been reported to undergo hepatic metabolism mediated by cytochrome P450 enzymes. This metabolic process produces a reactive metabolite capable of covalently binding to microsomal epoxide hydrolase, thereby triggering the formation of anti-microsomal epoxide hydrolase autoantibodies (Chen et al., 2011). Notably, a previous study has indicated that these autoantibodies can persist in the serum for a long period time (De Berardinis et al., 2000). Given their prolonged presence in the serum, anti-microsomal epoxide hydrolase autoantibodies have the potential to be promising biomarkers for assessing the hepatotoxicity of diosbulbin D. However, their clinical application requires further validation through comprehensive experimental studies and robust clinical evidences.

Studies have suggest that other proteins, such as beta 2-microglobulin, lipocalin-2, osteopontin, and cystatin C, may also serve as potential biomarkers for nephrotoxicity induced by traditional herbal medicines (Gao et al., 2023; Fuchs et al., 2014). These proteins listed above have been identified as potential biomarkers for assessing toxicity induced by TCM and its toxic compounds (Table 5). However, further more clinical evidences are needed to validate whether they can actually be used as biomarkers for TCM-induced toxicity.

Table 5
Proteins as biomarkers for toxicity induced by TCM.

Type	Protein	Trend	Reference
Cytokines	Interleukin-1 beta	↑	Tu et al. (2019); Tu et al. (2021)
	Interleukin-6	↑	
	Interleukin-10	↑	
	Interferon gamma	↑	
	Growth related oncogene alpha	↑	
	Granulocyte-macrophage colony-stimulating factor	↑	
	Tumor necrosis factor alpha	↑	
	Monocyte chemotactic protein-1	↑	
	Monocyte chemotactic protein-3	↑	
	Macrophage inflammatory protein-1 alpha	↑	
	Regulated upon activation normal T cell expressed and secreted	↑	
	Interferon gamma-inducible protein-10	↑	
	Vascular endothelial growth factor	↑	
	Cluster of differentiation 30 ligand	↑	Sheng et al. (2014)
	Interleukin-3	↓	
Other proteins	Kidney injury molecule-1	↑	Sabbisetti et al. (2013); Gao et al., 2023; Ebohon et al. (2020)
	Glycogen phosphorylase	↑	Wei et al. (2014)
	Glycine N-methyltransferase	↑	
	Cholinesterase	↑	Wang et al. (2017)
	Squalene synthase	↑	Li et al. (2023)
	Anti-microsomal epoxide hydrolase autoantibodies	↑	Chen et al. (2011)
	Beta 2-microglobulin	↑	Gu et al. (2024); Fuchs et al. (2014)
	Lipocalin-2	↑	
	Osteopontin	↑	
	Cystatin C	↑	

7. Conclusion

Herein, we have systematically discussed and summarized biomarkers for TCM-induced toxicity, and classified them in a logical and comprehensive manner. These biomarkers are characterized by their high sensitivity and specificity, making them suitable for be applied in clinic to reflect toxic damage caused by TCM.

Firstly, we explored the significant potential of DNA/protein adducts as biomarkers for TCM-induced toxicity. These adducts showed a high level of injury specificity and could accurately determine the type of damage induced by TCM. For example, pyrrole-protein adducts could indicate liver injury caused by TCM containing PA (Lin et al., 2011), while aristolactam-DNA adducts could reflect kidney damage induced by TCM containing AA (Jelakovic et al., 2012). However, there are still some shortcomings for DNA/protein adducts to be used as biomarkers, because they require precise instruments such as LC/MS for detection, resulting in the high testing cost. Furthermore, the measurement process is also susceptible to various factors, such as medication duration and potential interferences from other types of adducts. Additionally, the correlation between adducts and TCM-induced toxicity needs further validation through extensive clinical trials to ensure reliability and applicability.

Non-coding RNAs have emerged as potential early-warning biomarkers for TCM's toxicity due to their high sensitivity and tissue specificity. For instance, miRNA-122-3p has been reported to increase rapidly in liver injury induced by TCM containing diosbulbin-B, and its elevation occurs earlier than the change of classic biomarkers such as ALT and AST. Similarly, serum miRNA-5099 has been experimentally discovered as a specific biomarker for damage caused by diosbulbin-B (Yang et al., 2017b). Meanwhile, circulating non-coding RNAs can be detected using various RNA sequencing methods, which is more convenient. However, the current understanding of the biological functions and molecular mechanisms of non-coding RNAs remains limited. Therefore, it is necessary to clarify the importance of differential expression of non-coding RNAs in TCM's toxic reactions. What also cannot be ignored is that more effective, accurate, and feasible RNA detection methods should be developed to meet the requirements of medical diagnosis and to improve the early detection and management of TCM-induced toxicity.

Metabolites have the distinct advantage of serving as biomarkers because they can reflect pathological alterations in tissues or organs, and the metabolic characteristics of diseases can assess risk or provide specific disease subtypes. However, there are significant challenges that hinder the application of metabolites as new biomarkers for TCM-induced toxicity. The primary issue is the difficulty in the identification and validation of metabolites, largely due to the current limitations in metabolomics technology and the coverage of metabolomics is still restricted. On the other hand, metabolic changes are so complex, and currently there is no unified strategy or comprehensive analysis platform to analyze all metabolites of the entire phenotype. To address these limitations, it is essential to establish unified, standardized and universally recognized protocols. Considering the complexity and difficulty of high-throughput analysis, data integration and individual differences in metabolites, further research is required to fully realize the potential of metabolites as reliable biomarkers for assessing TCM-induced toxicity.

Other potential biomarkers, such as cytokines, KIM-1, and others, are also considered to exhibit high sensitivity as compared with classic biomarkers, offering promising prospects for further development as biomarkers for assessing toxicity induced by TCM. However, the biomarkers reviewed above still have their own shortcomings. It is evident that the identification and validation of novel early-warning biomarkers require more extensive experimental research and robust clinical data. Looking ahead, the development of appropriate combination diagnostic strategies and integrating multiple biomarkers for joint diagnosis may be the main development direction. For example, highly specific DNA/protein adducts could be applied to determine the type of damage

由中医引起的，同时高度敏感的非编码RNA可以被用来辅助指示损伤的严重程度。

中医具有独特的优势，包括应用广泛、疗效高、使用安全。然而，我们也需要更加关注中医相关的潜在毒性。传统的毒理学生物标志物往往缺乏特异性，因此不足以有效且迅速地指示中医引起的复杂毒性。因此，有必要寻找能够准确反映中医毒性的新的、敏感的、早期预警的生物标志物。这一进展将显著促进中医的合理和安全临床应用。未来，随着进一步深入研究，更适合且特异的生物标志物有望被开发并应用于临床实践，从而确保中医在临床中的疗效和安全性。

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利益冲突声明

作者声明，他们没有已知的可能影响本文报道工作的竞争性财务利益或个人关系。

数据可用性

本文描述的研究未使用任何数据。

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caused by TCM, while the highly sensitive non-coding RNAs could be assisted to indicate the severity of injury.

TCM exhibits unique advantages, including widespread applications, high efficiency, and safe in usage. However, we also need to pay more attentions to the potential toxicity associated with TCM. Traditional toxicological biomarkers often lack specificity, so they are insufficient to effectively and quickly indicate the complex toxicities caused by TCM. Therefore, it is necessary to seek new, sensitive, and early-warning biomarkers capable of accurately reflecting TCM-induced toxicity. Such advancement would significantly contribute to the rational and safe clinical application of TCM. In the future, more suitable and specific biomarkers are expected to be developed and applied in clinical practice with further in-depth studies, thereby ensuring both the efficacy and safety of TCM in clinic.

CRediT authorship contribution statement

Xinnan Gu: Writing – original draft, Investigation, Data curation. **Yu Zou:** Investigation, Data curation. **Zhenlin Huang:** Investigation, Funding acquisition. **Mengjuan Wei:** Writing – review & editing. **Lili Ji:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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